The structural limits of final human decision-making in an AI-assisted environment.

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1. Introduction

The speed at which artificial intelligence (AI) technology is being integrated into military operations far outpaces the ability of legal and ethical discourse to keep up. The dispute between Anthropic and the U.S. Department of Defense that erupted in February 2026 brought to the forefront the fundamental question of who should control AI technology. Anthropic established two "red lines" for its AI model Claude, that it should not be used for mass domestic surveillance or fully autonomous weapons, while the Department of Defense demanded unrestricted access for "all lawful purposes," resulting in a head-on collision of positions.

In this dispute, the U.S. government claimed that it "does not intend to create fully autonomous weapons that operate without human intervention." However, even if this is true and humans always serve as the final decision-makers, the question of how effective such a safeguard truly deserves scrutiny.

This paper critically examines whether the human's final decision-making authority can function as a substantive safeguard in AI-assisted decision-making environments, using the concept of "automation bias", well established in cognitive psychology as its central axis. The discussion begins with an everyday context (software development), expands to an extreme context (military operations), analyzes the Anthropic–Pentagon dispute as a case study, and concludes with proposals for institutional alternatives.

2. Background

Current large language models (LLMs) operate on the Transformer architecture. Unlike the human brain, which performs reasoning and learning simultaneously through recursive connections, this architecture computes from scratch within a given context window each time. It does not learn stably through real-time dialogue, and there exists a fundamental structural difference from human cognition.

Nevertheless, despite these technical limitations, the pace of LLM advancement has been extraordinary. Just one year ago, in March 2025, the Gemini 2.5 series had barely been announced; now it is already treated as an outdated model, having been superseded by several generations. In the face of such speed, laws and institutions are structurally destined to fall behind. It typically takes at least one year from problem identification to legislative proposal, yet in the AI field, one year is enough time for multiple technological generations to turn over.

One of the core concepts in the international discourse on the military use of AI is "human-in-the-loop." This refers to the principle that human approval is required before an AI system selects a target or initiates engagement. The contrasting concept is "human-on-the-loop," a structure in which AI acts autonomously while humans can intervene to halt the process.

In 2023, the UN General Assembly adopted a resolution urging regulatory discussions on autonomous weapons (152 in favor, 4 against), and the International Committee of the Red Cross (ICRC) recommended the adoption of legally binding rules on autonomous weapons. However, international consensus remains elusive, and many nations regard AI as a strategic asset, placing greater emphasis on development competition rather than regulation.

3. Structural Limits of Human Oversight

Automation bias refers to the cognitive tendency of humans to excessively trust the recommendations or outputs of automated systems, uncritically accepting them even in the presence of warning signs or contradictory information. This is not simply a matter of "laziness" but a systematic bias well documented in cognitive psychology.

According to a systematic literature review published in AI & Society in 2025, automation bias has emerged as a particularly serious concern when AI is deployed in high-risk domains such as healthcare, law, and public administration. The study emphasizes that automation bias does not operate in isolation but interacts with other cognitive biases, including automation complacency, forming complex relationships.

Even more noteworthy are the findings published in Scientific Reports. In this study, participants performed a medical diagnostic task with assistance from a biased AI, and after AI support was discontinued, they reproduced the same error patterns the AI had exhibited. In other words, a phenomenon of AI bias being "inherited" by humans was observed. This suggests that automation bias is not merely a phenomenon that occurs during AI use but can persistently influence human judgment even after AI use has ended.

Automation bias is already being widely observed in everyday contexts. Examining the behavior of developers using AI coding assistants such as Claude Code or Codex, many tend to commit AI-generated code without sufficient review. Because AI-generated code generally works, the rigor of review gradually weakens over time.

This demonstrates that the structure of "humans as final decision-makers" is already being undermined at the everyday level. Since code bugs rarely pose life-threatening consequences, such practices may not be perceived as significant problems. Some may argue that one should not equate a battlefield environment—where lives hang in the balance—with vibe coders sitting in an office scrolling Instagram while waiting for AI to finish its work. However, what if these cognitive patterns operate identically, or even more vulnerably, in life-or-death environments?

4. The "Final Decision-Maker" in Military Environments

The combat environment imposes conditions fundamentally different from everyday settings. First, there is extreme time pressure. In most cases, there is no time to cross-verify intelligence about enemy positions multiple times. Second, the constant threat to survival induces severe stress that degrades judgment. Third, the military's vertical chain of command structurally suppresses independent judgment among lower-ranking members.

The international survey experiment by Horowitz and Kahn (2024), involving 9,000 participants, directly addresses automation bias in national security contexts. According to this study, background knowledge of and trust in AI have a complex influence on the likelihood of automation bias, and notably, as the stakes of a decision increase, humans also exhibit a tendency toward "algorithm aversion", becoming more cautious about algorithms. However, whether this aversion can be maintained under the time pressure of actual combat is a separate question.

During my military service in my home country, I experienced a situation in which our not-so-welcome neighbors to the north launched a missile toward our territory, triggering an air raid alert and placing the military on emergency status. In that moment, everyone in the unit was in a state of extreme tension, and the majority facing this situation were young people around my age, barely into their twenties.

Based on this experience, the following hypothetical scenario comes to mind. If an AI system had provided the information that "enemy forces are massing in a specific zone," how critically would soldiers in a state of extreme tension have scrutinized that information? Would they have verified multiple times whether there were truly enemy combatants or civilians before employing their weapons? It is unrealistic to expect that

humans who cannot even properly review code in everyday settings would more carefully scrutinize AI's judgment under the extreme stress of life-or-death situations.

The military's vertical chain of command is a structural factor that further amplifies the risk of automation bias. Obedience to superiors' orders is the fundamental principle in the military, and the further down the rank, the weaker the autonomy in decision-making and the accompanying sense of responsibility. If AI provides information as an "assistive tool" and a superior issues orders based on that information, then from the perspective of the frontline soldier, the final executor, both the AI's information and the superior's orders become "things that must be followed."

A contribution in the Bulletin of the Atomic Scientists cites the testimony of Colonel Stanislav Petrov, who was on duty during the 1983 Soviet nuclear false alarm incident, describing the human response within the nuclear command structure as "highly conditioned, with almost no discretionary judgment." Under extremely tight time constraints, there was virtually no room for human reasoning and logic to operate; the personnel at the command center were closer to automated devices mechanically following checklists. If AI were added to this system, the space for humans to exercise independent judgment would likely shrink even further.

The 1983 Petrov incident is an important historical case in this context. When the Soviet early warning system detected a launch of American intercontinental ballistic missiles, Petrov judged the system's alert to be a false alarm and decided not to report it to his superiors. His judgment was correct, and this decision is widely recognized as having prevented a nuclear war. The critical point is that Petrov was able to make this judgment because, as an engineer who had participated in the development of the software, he possessed an intuitive understanding of the system's limitations. As the complexity of AI systems increases, expecting operators with this level of system comprehension will become increasingly difficult.

5. Case Analysis

In February 2026, the conflict between Anthropic and the U.S. Department of Defense erupted publicly. In a contract with the Department of Defense worth up to $200 million, Anthropic had stipulated two conditions: that its AI model Claude must not be used for mass domestic surveillance or fully autonomous weapons. The Department of Defense demanded unrestricted access for "all lawful purposes." When Anthropic refused, President Trump ordered all federal agencies to cease using Anthropic's technology, and Secretary of Defense Hegseth announced plans to designate Anthropic as a "supply chain risk."

The Department of Defense's official position in this dispute was clear. A DoD spokesperson stated that there was "no intent" to use AI for fully autonomous weapons or mass domestic surveillance, adding that doing so would already be illegal. However, what the Department of Defense wanted was for restrictions on specific use cases not to be written into the contract. The Pentagon's logic was that determinations of legality are the responsibility of the end user, The Department of Defense, not something that a company should pre-emptively restrict.

The government's logic carries a powerful basis in democratic legitimacy. In the democratic system that humanity has built over thousands of years, decisions regarding national security fall within the authority of the government elected by the people. The argument that a private company exercising veto power over a nation's military operations runs counter to this principle carries considerable persuasive force.

Yet Anthropic's position is not without basis. The Department of Defense "saying" it has no intention of certain uses and having use restrictions codified in a contract are entirely different levels of assurance. Government intentions can change with changes in administration, and restrictions not specified in a contract carry no legal binding force. As Anthropic's CEO Dario Amodei stated, concerns about uses that exceed the scope at which AI technology can perform safely and reliably are grounded in technological reality.

What is particularly notable is that this dispute precisely exposes the limits of the principle of "human final decision-making authority." The Department of Defense repeatedly asserts that humans make the final decisions, but as the automation bias research discussed above demonstrates, in environments where AI is used as an assistive tool, the human's "final decision" is at significant risk of becoming little more than a formal procedure that rubber-stamps the AI's output. This gap is precisely what Anthropic is concerned about.

This dispute raises a fundamental question that goes beyond the matter of an individual contract: where should the authority determine the scope of AI use reside in the AI era? Traditionally, the development of weapons technology was state led, with private companies playing the role of providing technology in response to government demands. In the AI era, however, this structure has been inverted: core technologies are developed by private companies, and the government purchases them. This shift in power dynamics is the essence of this dispute.

6. Conclusion

As examined in this paper, the principle that "humans are the final decision-makers" is theoretically sound but struggles to function as a substantive safeguard in real operational environments due to the cognitive limitation of automation bias. This limitation is already

observable in everyday contexts (code review) and is expected to intensify in military contexts where time pressure, threats to survival, and vertical chains of command converge.

The dispute between Anthropic and the U.S. Department of Defense demonstrates that this issue is not confined to abstract discussion but is directly connected to real-world policy decisions. Rather than resting on the linguistic safeguard of "human final decision-making authority," we must acknowledge the structural conditions that make automation bias inevitable and design institutional mechanisms to mitigate it.

Specifically, the following directions are proposed. First, legislation on the military use of AI is urgent. As Anthropic's CEO requested of Congress, legal restrictions on mass surveillance and autonomous weapons should not rely on the conscience of AI-manufacturing companies and their self-imposed policies alone but must be established in the form of law.

In 1983, Colonel Petrov's ability to override the system's warning and exercise independent judgment was a miracle, but we must not assume that everyone in the military would make the same decision as Colonel Petrov. In an era where AI develops faster than the pace of social mobility, we should not leave matters to tacit agreement but instead deploy clear restrictions and use AI for clearly defined objectives.

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